

MCCB: Multi-Chamber-Camera Bioreactor

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MCCB Overview

The presence of electric and magnetic fields has been proved to influence **stem cell differentiation** interesting many researchers of regenerative medicine

Goal: Subject 4 different culture wells to various **electric and magnetic stimuli intensities** as a tool for bioengineers to study the different effects of the applied stimuli.

Design Overview: The team has designed an apparatus that allows for **consistent adjustable stimulation**, data collection, and **imaging** via a camera microscope.

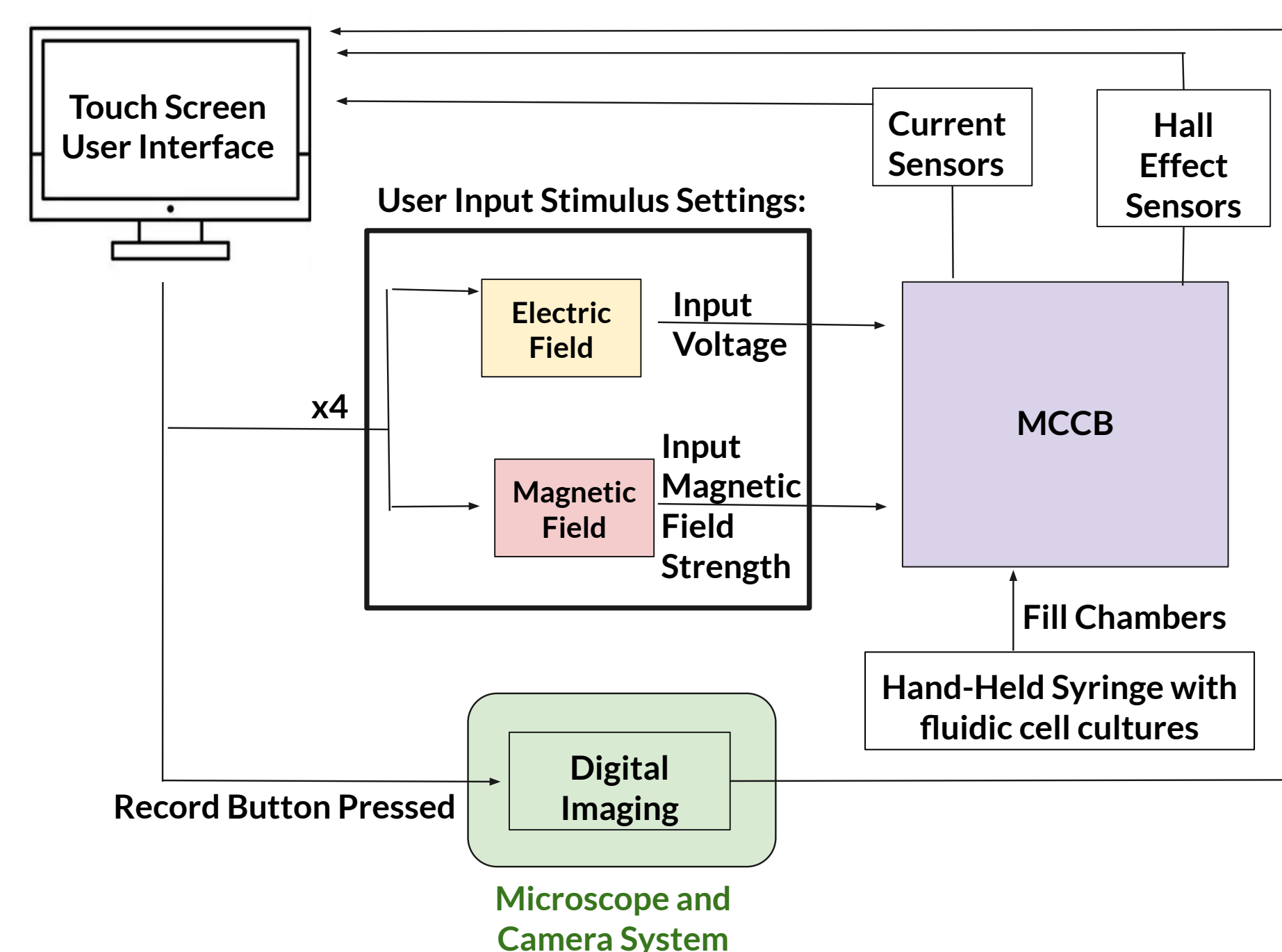


Figure 1: Operational Block Diagram

Central Electronics Overview

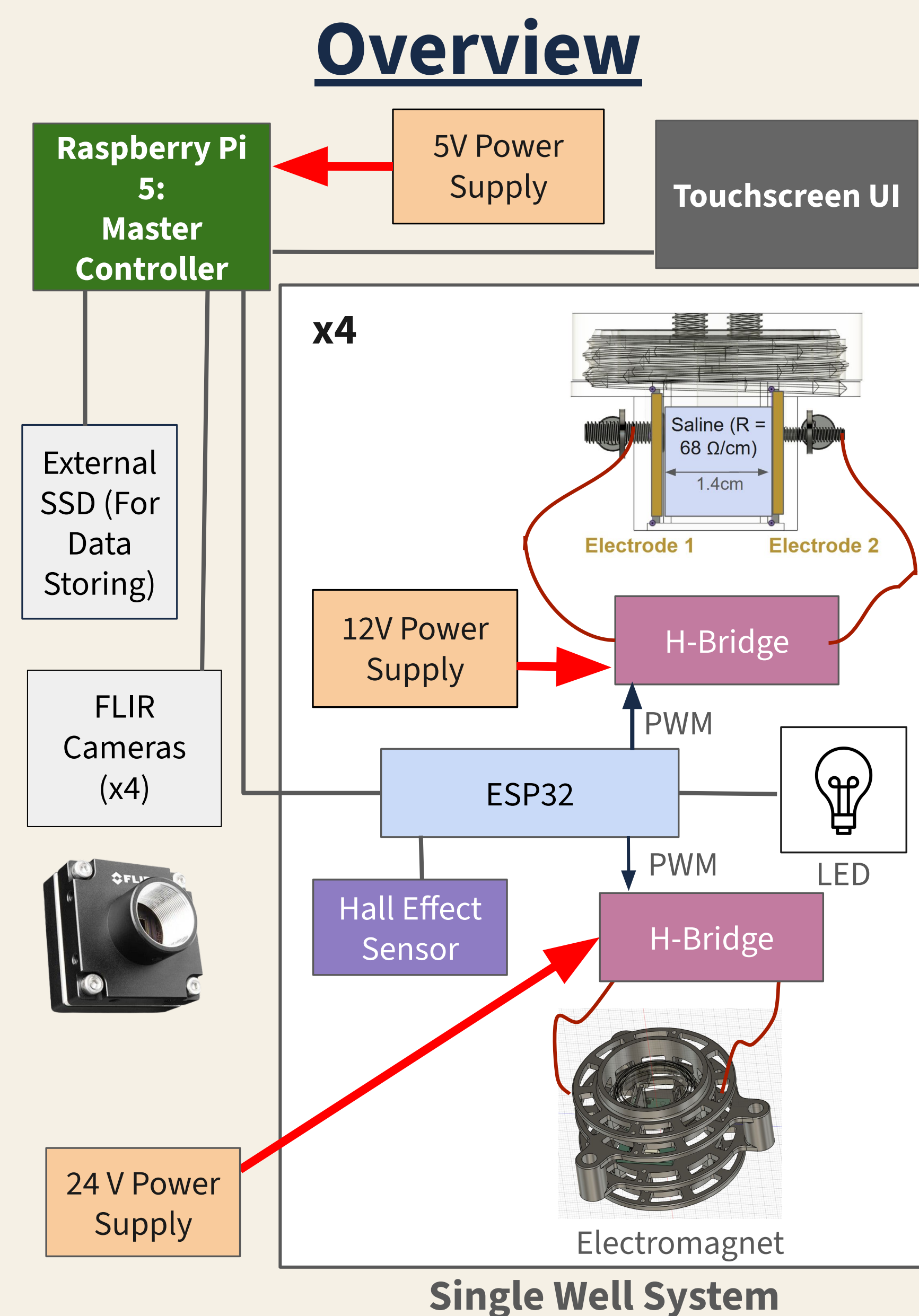


Figure 2: High Level Control System Layout

Integrated System

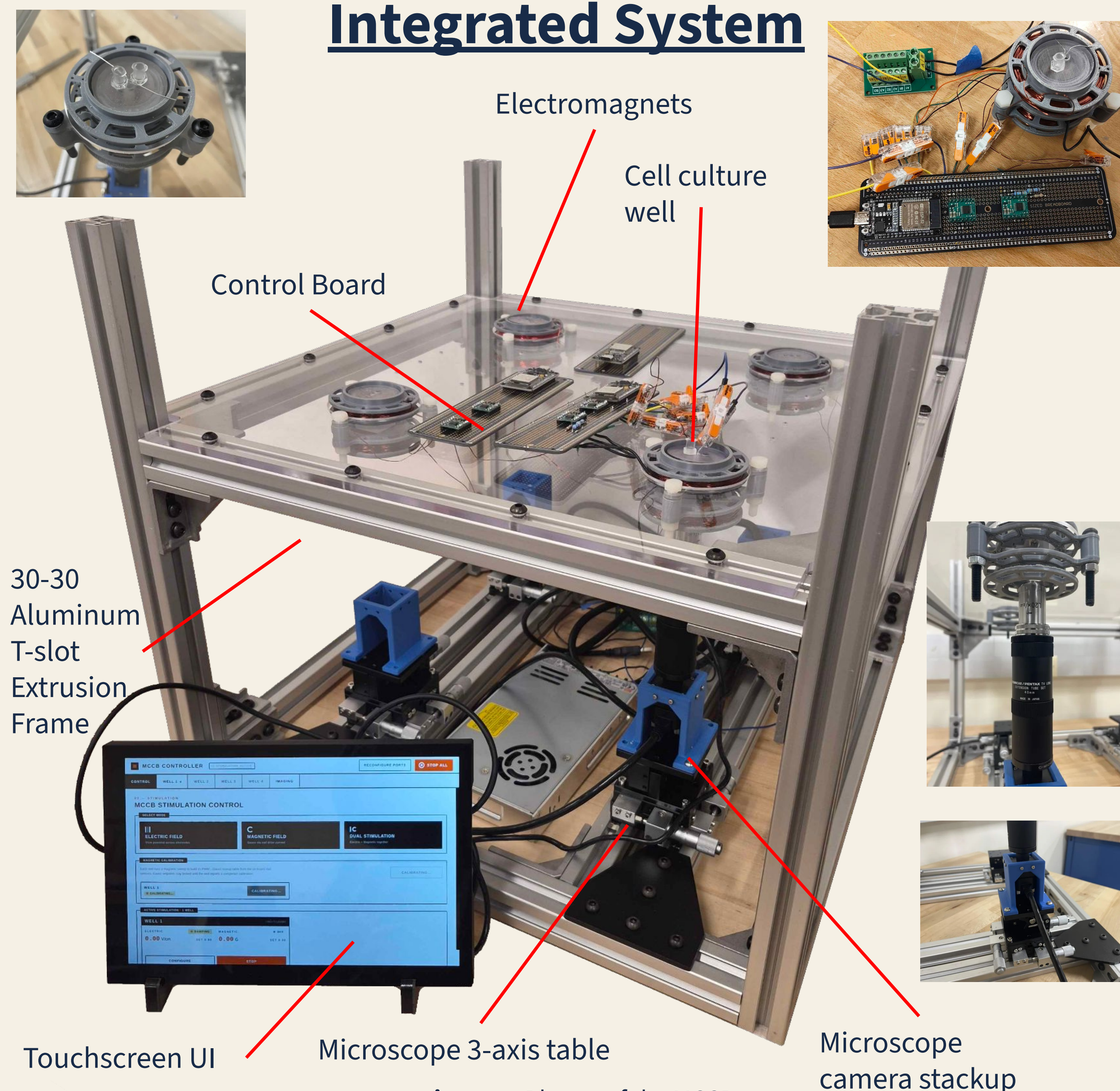
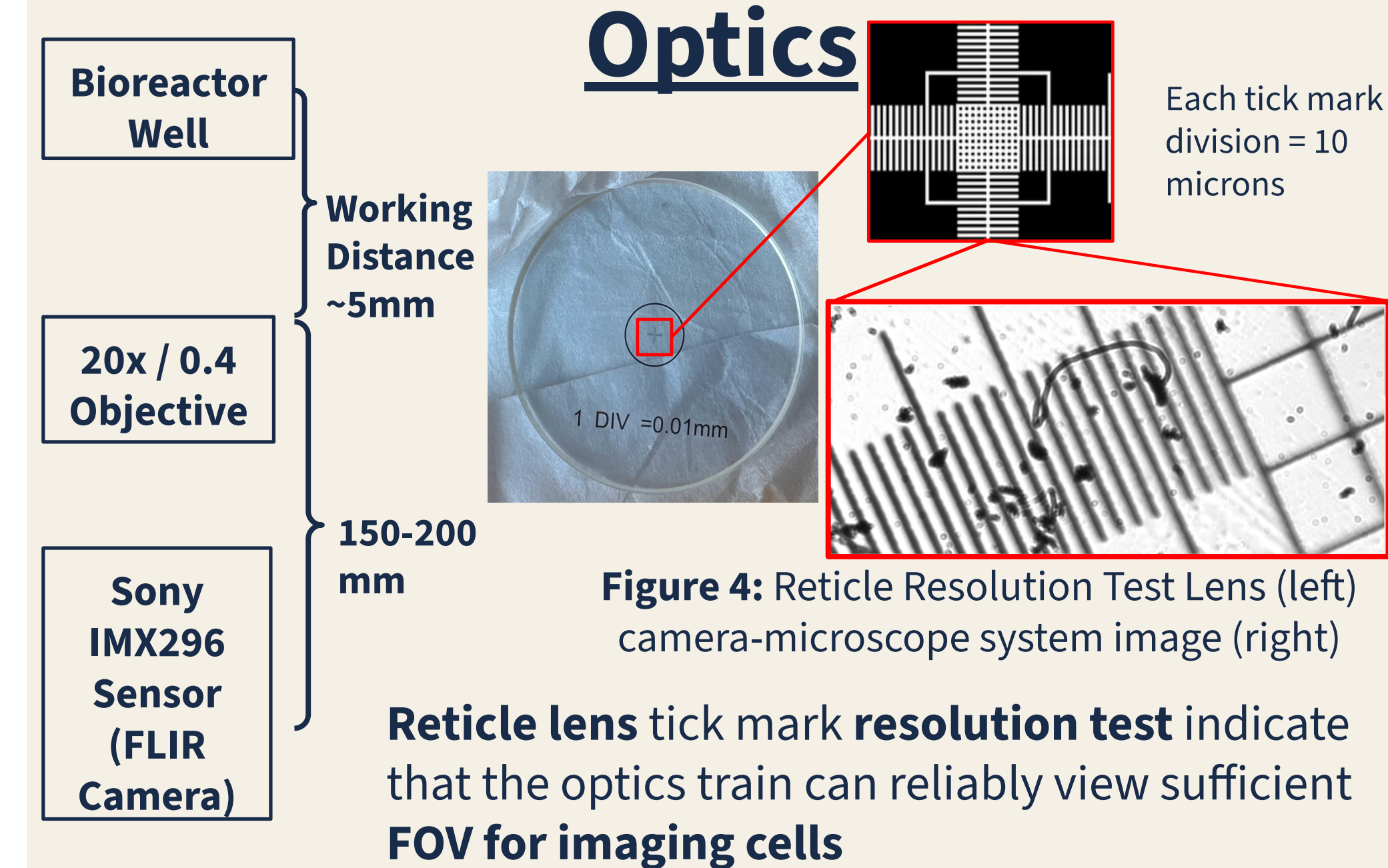


Figure 3: Photos of the MCCB

Optics



Well Design

The wells are printed from Veroclear Resin and sterilized in a chemical bath. Features such as the O-ring groove and threads are designed for 3D printing ensure a watertight seal. A central plunger allows for clearer cell observation and there are electrode slots within the chamber.

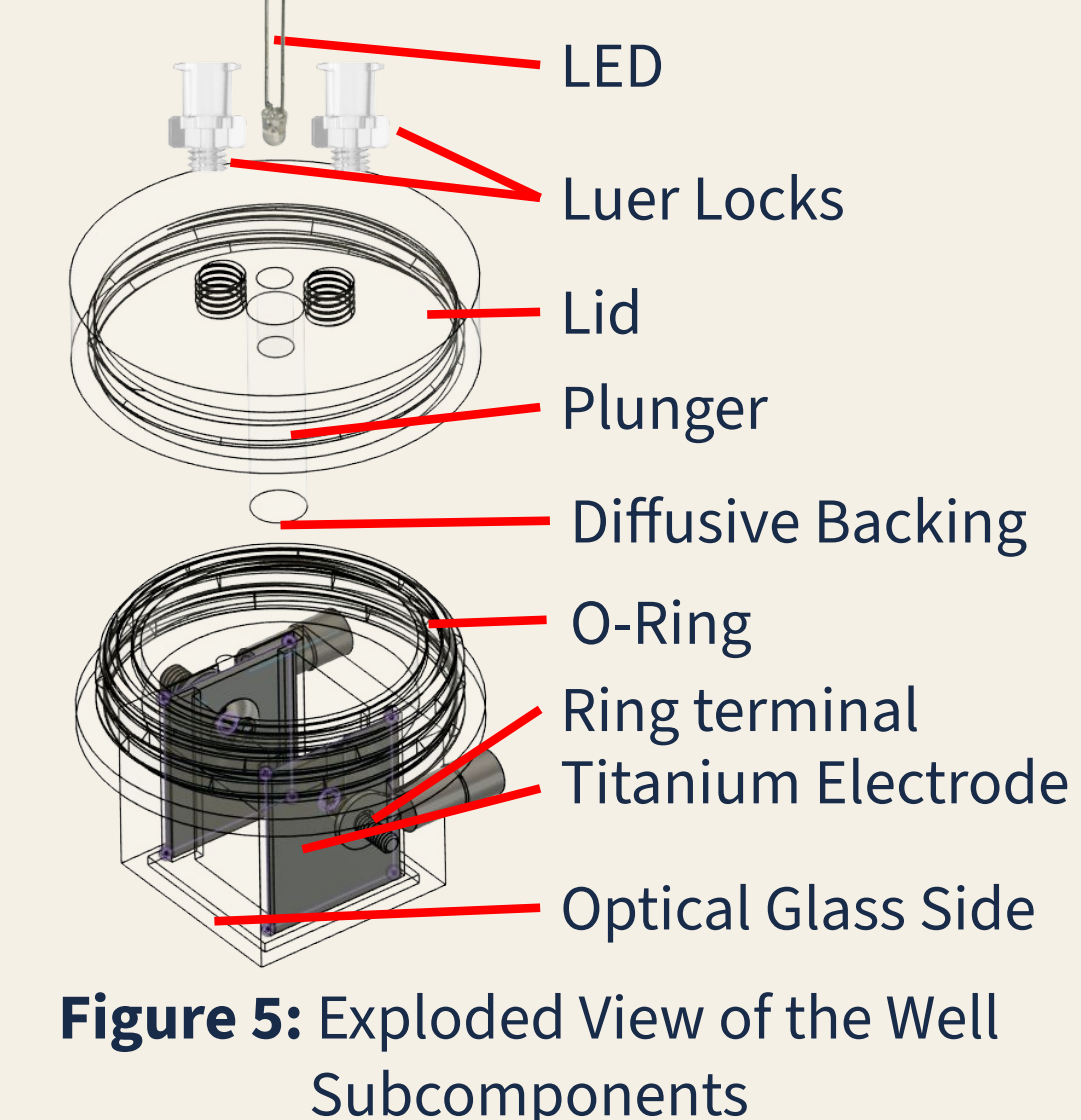
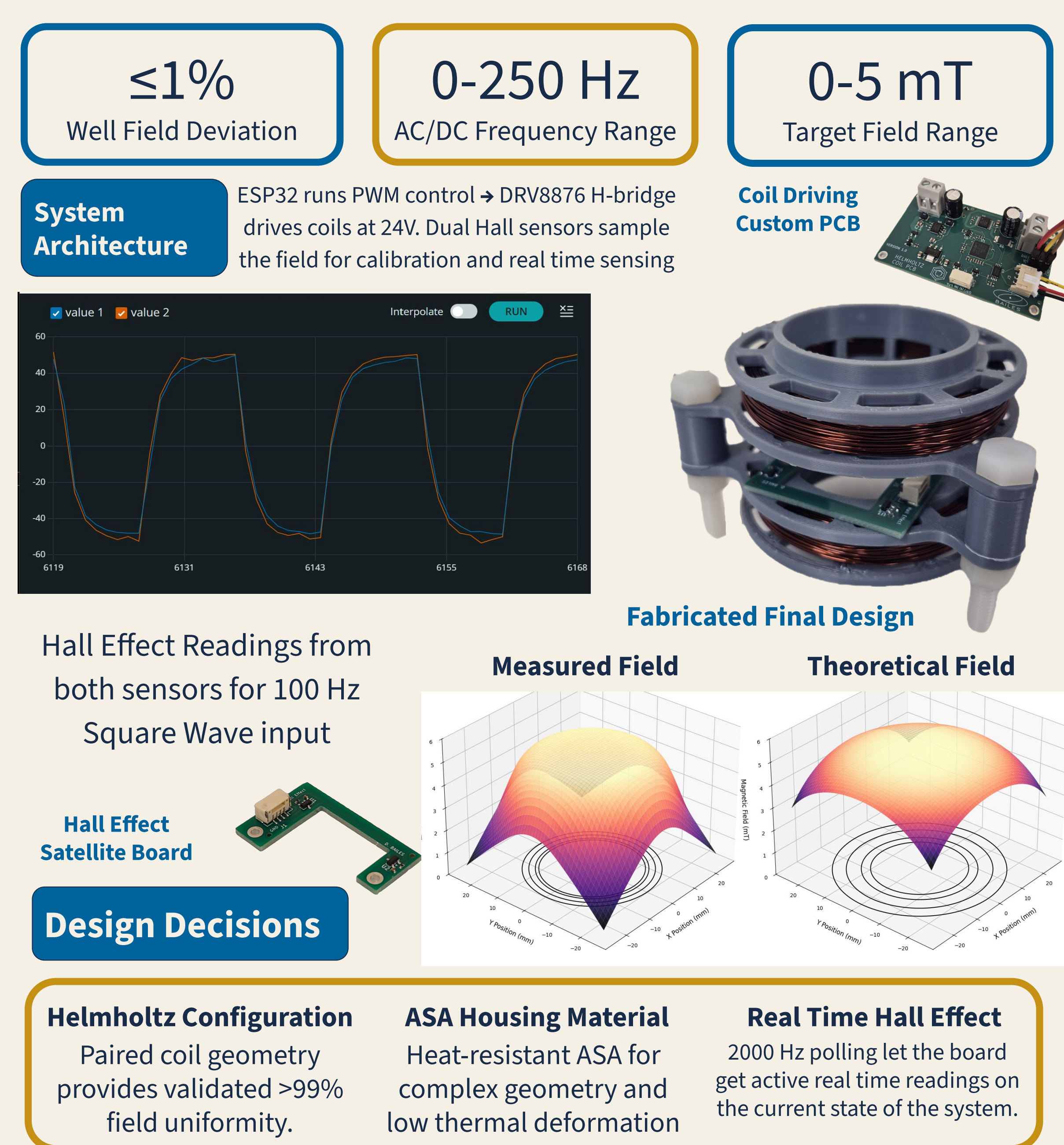
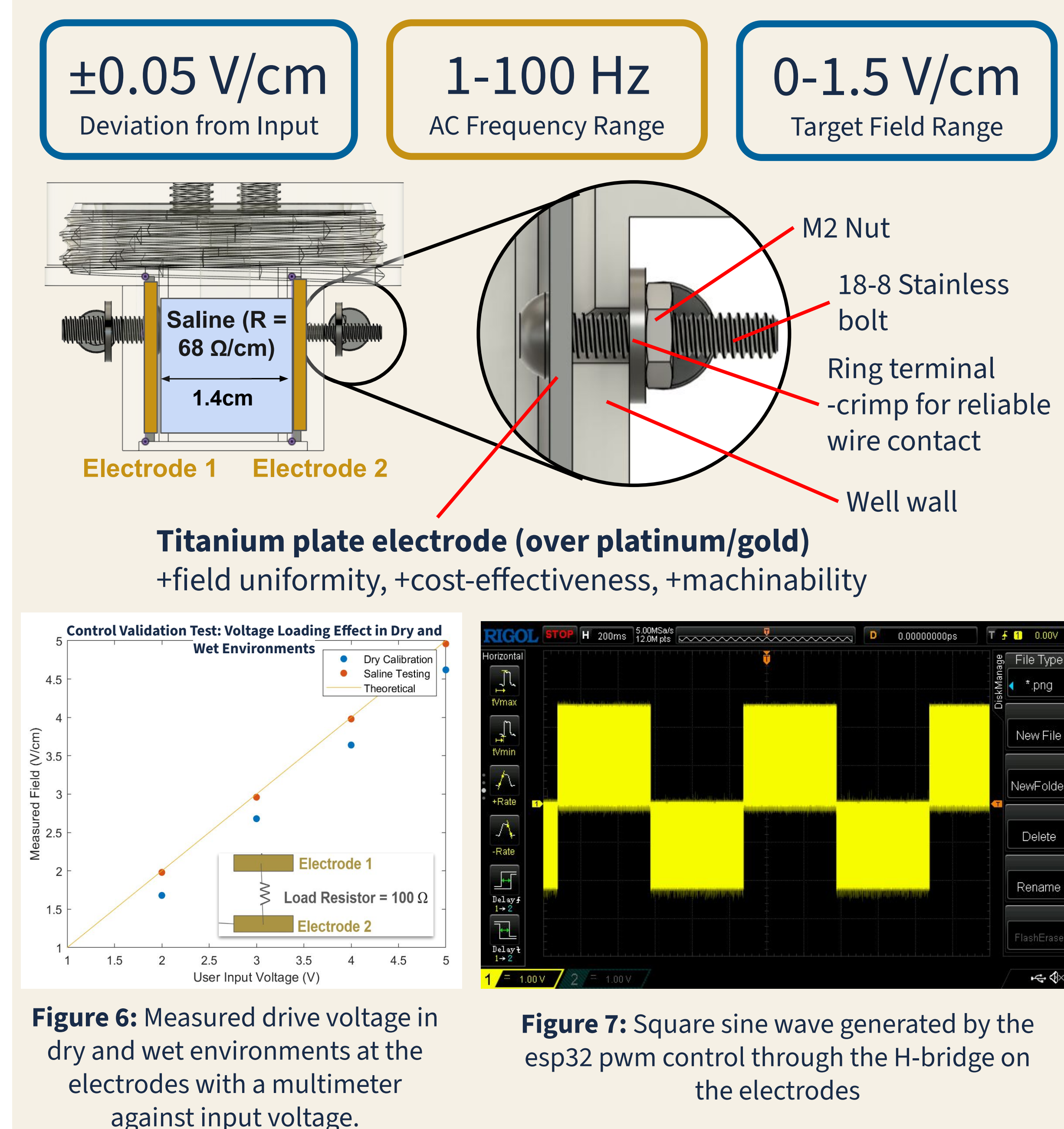


Figure 5: Exploded View of the Well Subcomponents

Electromagnetic Field Generation



Electric Field Generation



User Control

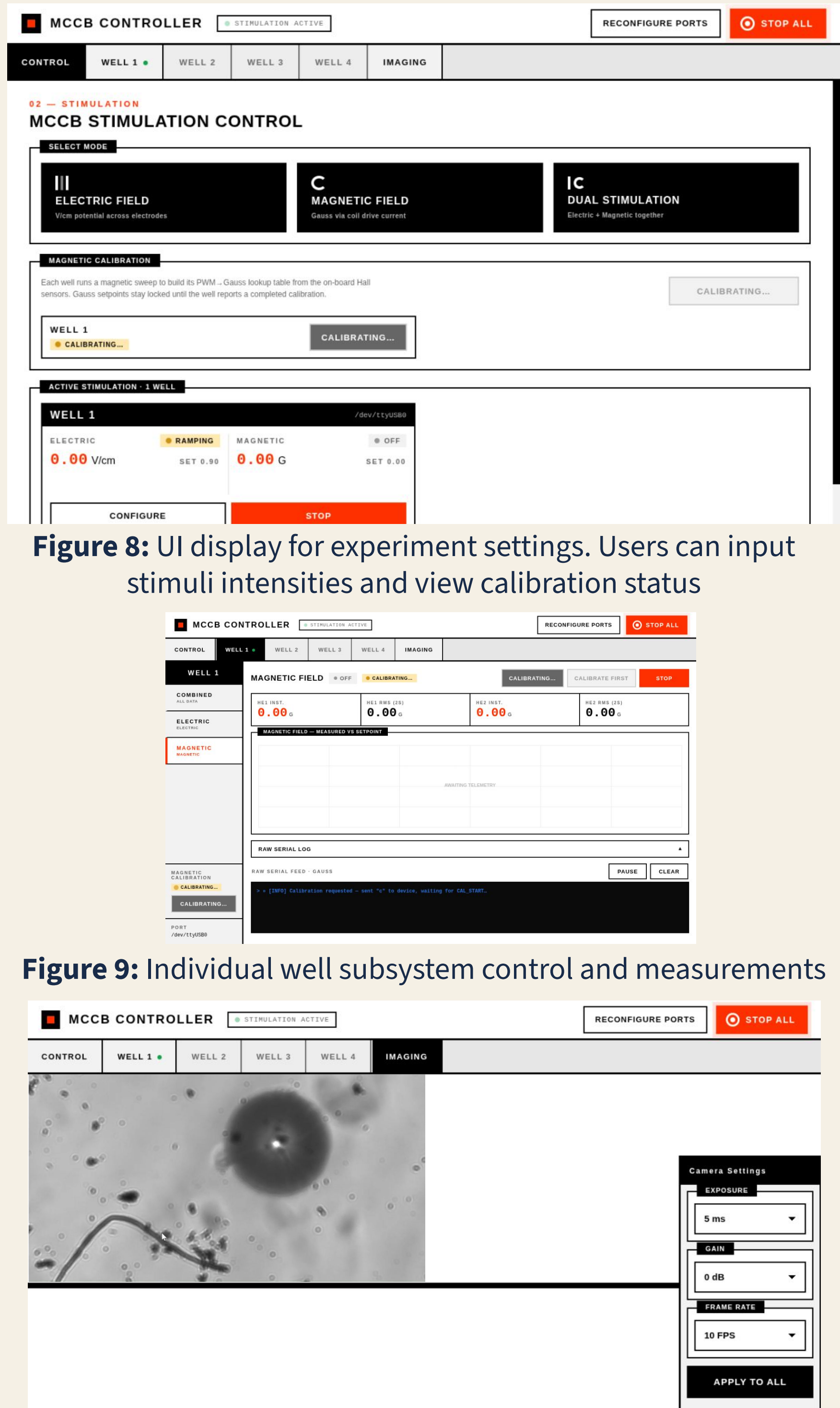


Figure 8: UI display for experiment settings. Users can input stimuli intensities and view calibration status

Design Recommendations

- Biocompatible Resin
- Printed Circuit Board Incorporation
- Downsize System Footprint
- Platinum Electrodes
- Improve Optics
- Incorporate Hydrostatic Pressure

Impact on Society

Expansion of **cell differentiation research** which could be impactful in the broader field of regenerative medicine.

Acknowledgements

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References

- [1] Martin, Ivan, David Wendt, Michael Heberer. The role of bioreactors in tissue engineering. Vol. 22. Issue 2. February 2004. [https://www.cell.com/trends/biotechnology/fulltext/S0157-7799\(03\)00318-4?ccy](https://www.cell.com/trends/biotechnology/fulltext/S0157-7799(03)00318-4?ccy)
- [2] Meng, Shiyun, Mahmoud Roushnia, Ze Zhang. Electrical Stimulation and Cellular Behavior in Biomedical Research. Editor Mieczyslaw Lapkowski. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8748014/>
- [3] Altamirano M., Coates C.W., Grundfest H., Nachmansohn D. Mechanisms of bioelectric activity in electric tissue. I. The response to indirect and direct stimulation of electroplaques of Electrophorus electricus. J. Gen. Physiol. 1953;37:91-110. doi: 10.1085/jgp.37.1.91
- [4] Neuffer, Chester B. Helmholtz Coils for Producing Uniform Magnetic Fields. Journal of the Optical Society of America, Vol. 13. Issue 2. pp. 205-212, 1926. <https://opg.optica.org/fulltext.cfm?uri=josa-13-2-205>